

Unit 13, Lesson 12: Congruence and Similarity



Benchmark

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to justify congruence and similarity in triangles and quadrilaterals.



13.12.1 Warm-Up: Notice and Wonder

Brickley, the LEGO sea serpent in Lake Buena Vista at Disney Springs near Orlando, Florida, was made in 1997 from 170,000 LEGO bricks and is 30 feet long.



1. What do you notice?
2. What do you wonder?



13.12.2 Exploration: Congruence and Similarity

1. Parallelogram $DKTS$ has vertices at $D(-2,4)$, $K(-4,1)$, $T(-9,3)$, and $S(-7,6)$.

Complete the table.

Line Segment	Length
$\overline{DK}$	
$\overline{KT}$	
$\overline{TS}$	
$\overline{SD}$	

2. Parallelogram $WRCN$ has vertices at $W(0,1)$, $R(2,-2)$, $C(7,0)$, and $N(5,3)$.

Complete the table.

Line Segment	Length
$\overline{WR}$	
$\overline{RC}$	
$\overline{CN}$	
$\overline{NW}$	

3. What do you notice about the lengths of the line segments in parallelograms $DKTS$ and $WRCN$?

4. Complete the statement.

Parallelogram $DKTS$ may be

- ☐ congruent
- ☐ similar

to parallelogram $WRCN$ because all corresponding sides

are

- ☐ proportional.
- ☐ congruent.

5. Work with your partner to describe how the parallelograms are congruent or similar using transformations. State which image you used as the preimage.



6. Isosceles trapezoid $MPHN$ has vertices at $M(2, -6)$, $P(3, -3)$, $H(5, -2)$, and $N(8, -3)$.

Complete the table.

Line Segment	Length
$\overline{MP}$	
$\overline{PH}$	
$\overline{HN}$	
$\overline{NM}$	

7. Isosceles trapezoid $WVYZ$ has vertices at $W(4, 12)$, $V(6, 6)$, $Y(10, 4)$, and $Z(16, 6)$.

Complete the table.

Line Segment	Length
$\overline{WV}$	
$\overline{VY}$	
$\overline{YZ}$	
$\overline{ZW}$	

8. What do you notice about the lengths of the line segments in isosceles trapezoids $MPHN$ and $WVYZ$?

9. Complete the statement.

Isosceles trapezoid $MPHN$ is
☐ congruent
☐ similar
 to

isosceles trapezoid $WVYZ$ because all corresponding

sides are
☐ proportional.
☐ congruent.

10. Work with your partner to describe how the trapezoids are congruent or similar using transformations. State which image you used as the preimage.



13.12.3 Guided Practice: Congruence and Similarity

1. Right triangle CKR has vertices at $C(-2,5)$, $K(-8,5)$, and $R(-8,2)$. Right triangle DPM has vertices at $D(-2,-4)$, $P(4,-4)$, and $M(4,-1)$.
 - a. Determine if $\triangle CKR$ and $\triangle DPM$ are congruent, similar, or neither.
 - b. What definition, theorem, or property justifies your conclusions?
 - c. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the preimage.

- d. Ask a classmate for their description. Record their description and write your summary of their description. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Description	My Summary of Their Description	Initials

2. Parallelogram $HGNC$ has vertices at $H(5, -1)$, $G(2, 3)$, $N(9, 3)$, and $C(12, -1)$. Parallelogram $VYZW$ has vertices at $V(-1, 1.5)$, $Y(0.5, 4)$, $Z(-3, 4)$, and $W(-4.5, 1.5)$.

- a. Shin and Kendal are determining if parallelograms $HGNC$ and $VYZW$ are congruent, similar, or neither. Their work is shown.

Shin's Work	Kendal's Work
Parallelograms $HGNC$ and $VYZW$ are similar because $\frac{ZY}{NG} = \frac{VW}{HC}$	Parallelograms $HGNC$ and $VYZW$ are neither because $\frac{ZY}{NG} \neq \frac{YV}{GH}$

- b. Whose work do you agree with?
- c. Write a note to the person who is incorrect explaining their mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills



13.12.4 Your Turn!

Complete the table by determining if the given polygons are congruent, similar, or neither.

1. $\triangle STK$: $S(-6,8)$, $T(-10,5)$, and $K(-6,1)$

$\triangle NRC$: $N(3,7.5)$, $R(-3,3)$, and $C(3,-3)$

- ☐ Congruent
- ☐ Similar
- ☐ Neither

2. Trapezoid $MPWY$: $M(3,4)$, $P(7,6)$, $W(10,5)$, and $Y(4,2)$

Trapezoid $LKGH$: $L(-2,6)$, $K(-8,9)$, $G(-14,7)$, and $H(-4,2)$

- ☐ Congruent
- ☐ Similar
- ☐ Neither

3. Kite $XZWY$: $X(-7,6)$, $Z(-6,9)$, $W(-3,10)$, and $Y(-1,4)$

Kite $FGDK$: $F(2,3)$, $G(1,6)$, $D(-2,7)$, and $K(-4,1)$

- ☐ Congruent
- ☐ Similar
- ☐ Neither



13.12.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to teach them how to use coordinate geometry to justify congruence and similarity in polygons.

Unit 13, Lesson 13: Finding Perimeter and Area on the Coordinate Plane



Benchmark

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving the perimeter or area of polygons.



Learning Target

- ☐ I can find the perimeter and area of polygons and composite figures drawn on the coordinate plane.